Chapter 4, Problem 72P

Problem

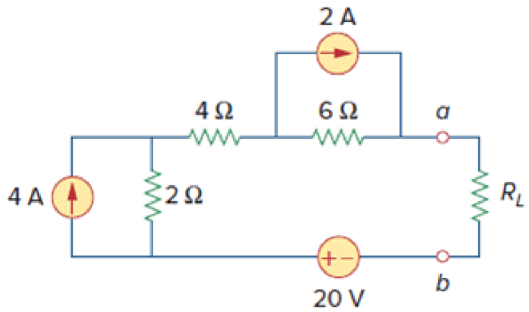
(a) For the circuit in Fig. 4.138, obtain the Thevenin equivalent at terminals *a*-*b*.

(b) Calculate the current in *RL* = 13 Ω.

(c) Find *RL* for maximum power deliverable to *RL*.

(d) Determine that maximum power.

Figure. 4.138:



Step-by-step solution

Step 1 of 9

Refer to Figure 4.138 in the text book.

(a)

Disconnect the load resistance to calculate the Thevenin's voltage.

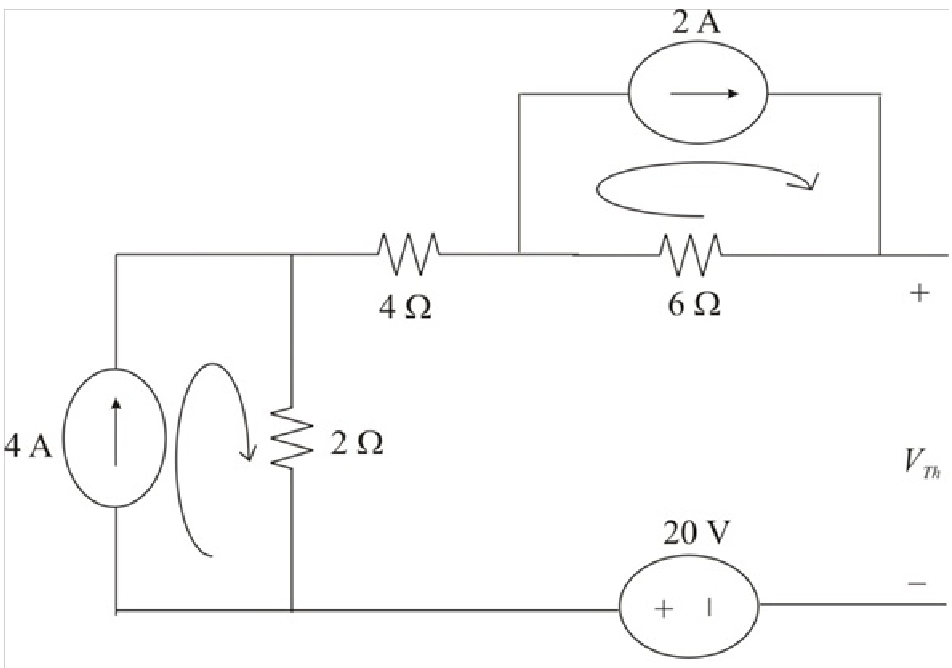


Figure 1

Comments (1)

Anonymous

Why there is no current in 4ohms? and how come 20V is positive?

Step 2 of 9

Notice that no current flows through $4\ \Omega$ resistor.

Calculate the value of the Thevenin's voltage V_{Th}.

$$\begin{aligned} V_{Th} &= (2)(6)+(4)(2)+20 \\ &= 12+8+20 \\ &= 40\text{ V} \end{aligned}$$

Therefore, the value of the Thevenin's voltage V_{Th} is 40 V.

Step 3 of 9

Set the values of the sources to calculate the value of the Thevenin's resistance.

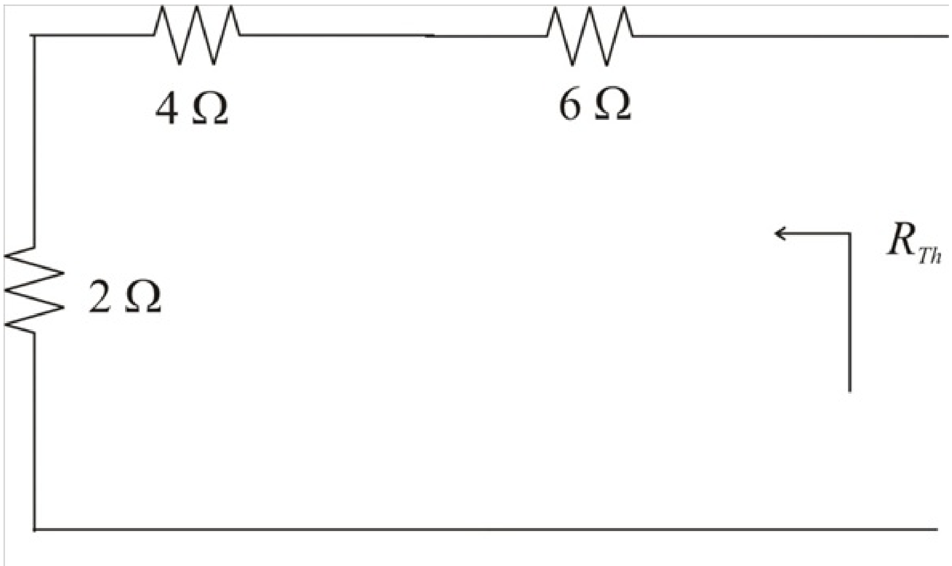


Figure 2

Step 4 of 9

Calculate the value of the Thevenin's resistance R_{Th}.

$$\begin{aligned} R_{Th} &= 6+4+2 \\ &= 12\ \Omega \end{aligned}$$

Therefore, the value of the Thevenin's resistance R_{Th} is $12\ \Omega$.

Step 5 of 9

Draw the Thevenin's equivalent circuit diagram.

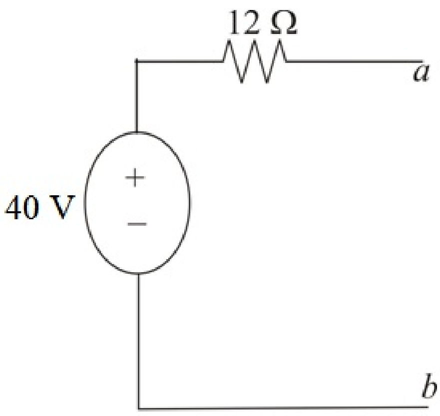


Figure 3

Step 6 of 9

(b)

Connect the load resistance to the Thevenin's circuit as show in Figure 4.

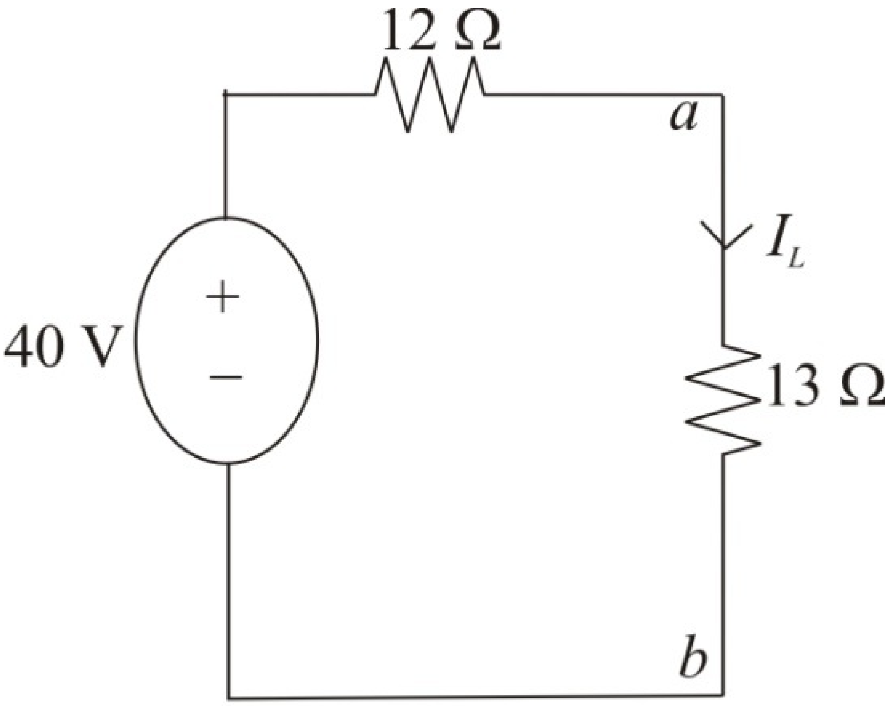


Figure 4

Step 7 of 9

Calculate the value of the load current I_L.

$$
\begin{aligned}
I_L &= \frac{40}{12+13} \\
&= \frac{40}{25} \\
&= 1.6 \text{ A}
\end{aligned}
$$

Therefore, the value of the load resistance I_L is $\boxed{1.6 \text{ A}}$.

Step 8 of 9

(c)

For the maximum power to be delivered to the load, the value of the load must be equal to the Thevenin's resistance R_{Th}. Therefore, the value of the load R_L is $\boxed{12\ \Omega}$.

Step 9 of 9

(d)

Calculate the maximum power absorbed by the load.

$$p_{\max} = \frac{V_{Th}^{\ 2}}{4R_L}$$

Substitute $40\ \text{V}$ for V_{Th} and $12\ \Omega$ for R_L.

$$\begin{aligned} p_{\max} &= \frac{(40)^2}{4(12)} \\ &= \frac{1600}{48} \\ &= 33.33\ \text{W} \end{aligned}$$

Therefore, the value of the maximum power absorbed by the load is $\boxed{33.33\ \text{W}}$.